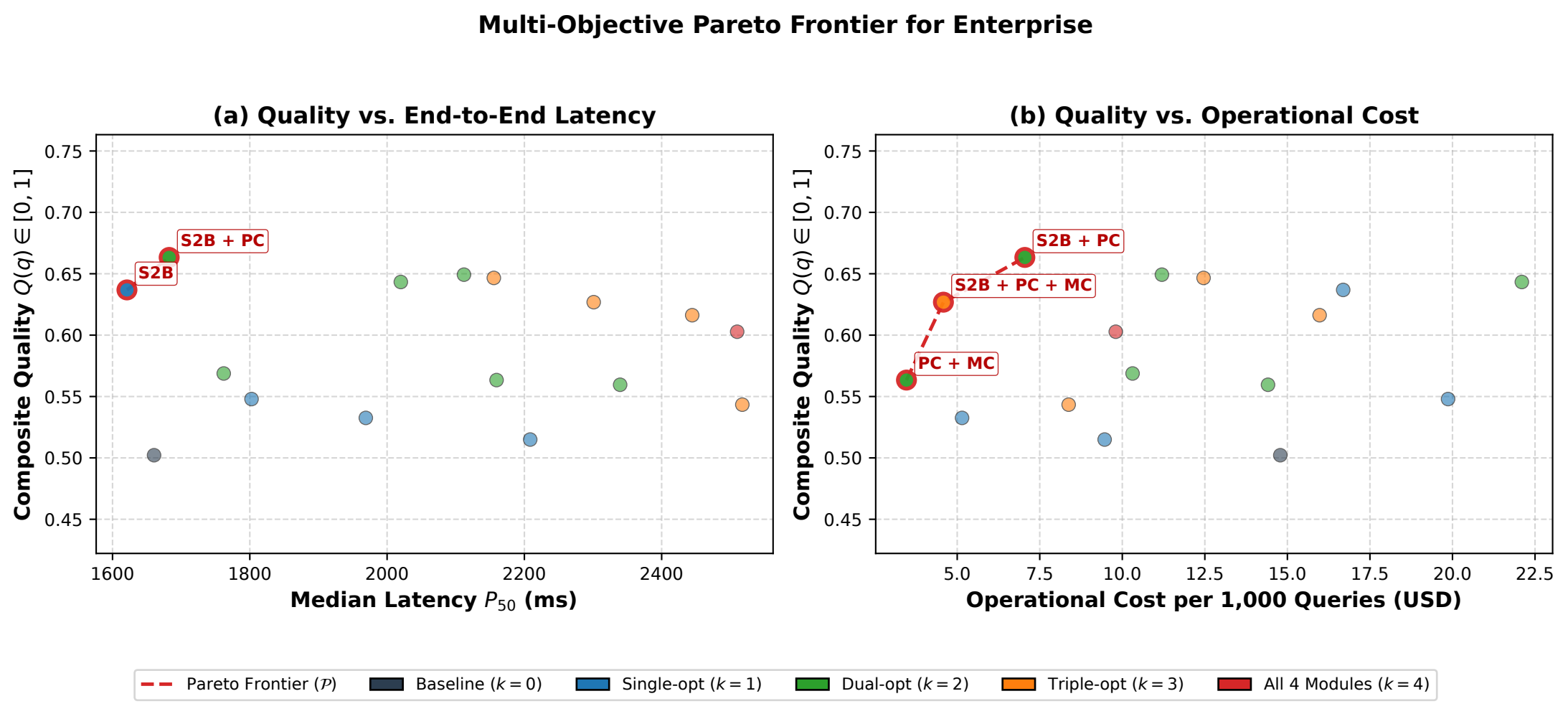
Multi-Objective Pareto Frontier for Enterprise
(a) Quality vs. End-to-End Latency
(b) Quality vs. Operational Cost
Composite Quality $Q(q) \in [0, 1]$
Median Latency P_{50} (ms)
Operational Cost per 1,000 Queries (USD)
S2B
S2B + PC
S2B + PC + MC
PC + MC
0.75
0.70
0.65
0.60
0.55
0.50
0.45
1600
1800
2000
2200
2400
5.0
7.5
10.0
12.5
15.0
17.5
20.0
22.5
Pareto Frontier ($\mathcal{P}$)
Baseline ($k = 0$)
Single-opt ($k = 1$)
Dual-opt ($k = 2$)
Triple-opt ($k = 3$)
All 4 Modules ($k = 4$)